Practical Worksheet 12.03.2026

Create a database Part_shipment.

Create the following tables in the **Part_shipment** database:

```
create table suppliers (S_ID varchar(6) not null primary key,
S_Name char(25),
Status int(3),
City char(25));

create table Parts (P_ID varchar(6) not null primary key,
P_Name char(25),
Colour char(25),
Weight int(3),
City char(25));

create table Project (Pr_ID varchar(6) not null primary key,
Pr_Name char(25),
City char(25));

create table Shipment (S_ID varchar(6) not null,
P_ID varchar(6) not null,
Pr_ID varchar(6) not null,
Quantity int(4),
primary key(S_ID, P_ID, Pr_ID),
foreign key(S_ID) references suppliers(S_ID),
foreign key(P_ID) references Parts(P_ID),
foreign key(Pr_ID) references Project(Pr_ID));
```

```
insert into suppliers(S_ID,S_Name,Status,City) values
('S1','Smith',20,'London'),
('S2','Jones',10,'Paris'),
('S3','Black',30,'Paris'),
('S4','Clark',20,'London'),
('S5','Adams',30,'Athens');


insert into parts(P_ID,P_Name,Colour,Weight,City) values
('P1','Nut','Red',12,'London'),
('P2','Bolt','Green',17,'Paris'),
('P3','Screw','Blue',17,'Rome'),
('P4','Screw','Red',14,'London'),
('P5','Cam','Blue',12,'Paris'),
('P6','Cog','Red',19,'London');


insert into project(Pr_ID,Pr_Name,City) values
('J1','Sorter','Paris'),
('J2','Punch','Rome'),
('J3','Reader','Athens'),
('J4','Console','Athens'),
('J5','Collator','London'),
('J6','Terminal','Oslo'),
('J7','Tape','London');
```

```
insert into Shipment(S_ID,P_ID,Pr_ID,Quantity) values
('S1','P1','J1',200),
('S1','P1','J4',700),
('S2','P2','J1',400),
('S2','P3','J2',200),
('S2','P3','J3',200),
('S2','P3','J4',500),
('S2','P3','J5',600),
('S2','P3','J6',400),
('S2','P3','J7',800),
('S2','P5','J2',100),
('S3','P3','J1',200),
('S3','P4','J2',500),
('S4','P6','J3',300),
('S4','P6','J7',300),
('S5','P2','J2',200),
('S5','P2','J4',100),
('S5','P5','J5',500),
('S5','P5','J7',100),
('S5','P6','J2',200),
('S5','P1','J4',100),
('S5','P3','J4',200),
('S5','P4','J4',800),
('S5','P5','J4',400),
('S5','P6','J4',500);
```

List down the index of each of the above tables by executing the following MySQL statement:

Syntax: `SHOW INDEX FROM <table name>;`

Eg: `SHOW INDEX FROM suppliers;`

Creating Indexes

Syntax for index creation:

```
CREATE INDEX index_name ON table_name (column1);
```

Syntax for Composite index creation:

```
CREATE INDEX index_name on table_name (column1, column2);
```

Syntax for deleting index:

```
DROP INDEX index_name ON table_name;
```

Answer the following questions on indexes:

1. Create an index on Quantity of the Shipment table.
2. List down the indexes from Shipment table.
3. Execute the following MySQL statement:
   `explain select * from shipment where Quantity = 500;`
4. Create an index on S_Name on the Suppliers table
5. Create a composite index on P_Name and Colour of the Shipment table.
6. Delete the index created in Question 1.

Exercise:

Consider the Hospital database created in the last practical and Create a single-column index and a composite index for each of the tables.